

# Training Needs Analysis Automation

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## ABSTRACT

This report presents the Training Needs Analysis Automation project, a web-based system developed to make employee training need detection more organized, measurable and data-driven. The system manages users, employees, evaluations, competency scores, training recommendations, training plans and participation records through a single platform.

Airtable is used as the database layer, Next.js is used for the user interface, Clerk is used for authentication, Vercel is planned for deployment, and n8n with OpenRouter is prepared for AI-supported automation. The main contribution of the project is reducing manual review work for human resources teams and turning evaluation results into practical training actions.

Keywords - training needs analysis, human resources, automation, artificial intelligence, Next.js, Airtable, n8n, Clerk, OpenRouter

## I. INTRODUCTION

Organizations need to follow employee skills and competency levels regularly in order to plan effective training activities. In manual systems, evaluation results are often kept in separate files, and training needs are identified by reviewing many records one by one. This creates extra work for human resources staff and makes the process difficult to control.

The developed project solves this problem by collecting the related steps in one digital system. It is not only a database display panel; it is a usable application where users can log in, work according to their roles, create employee evaluations, review training recommendations, prepare training plans and follow participation status.

The application is designed especially for the problems of data disorder, repetitive manual checks and reporting difficulty in training need analysis. By connecting evaluation data with recommendation and planning steps, the process becomes more traceable and measurable.

## II. PURPOSE AND SCOPE OF THE PROJECT

The main purpose of the project is to determine employee training needs from evaluation data. The system identifies low-scoring competency areas, connects them with suitable training suggestions and supports the decision-making process of the human resources team.

The scope includes storing user and employee information, creating competency-based evaluations, generating training recommendations, converting approved recommendations into training plans and allowing employees to follow their own training process. The system is also prepared for automatic recommendation generation with n8n and OpenRouter.

## III. SYSTEM ARCHITECTURE

The project uses a modular architecture. Airtable is the data source, Next.js provides the frontend layer, Clerk manages authentication and sessions, n8n is prepared for background workflows, and OpenRouter is selected for the artificial intelligence model connection. This structure makes the system easier to manage during development and easier to extend later.

When a user signs in, Clerk identifies the account. The application reads the related user record from Airtable and receives the role information. Then the sidebar and page permissions are arranged according to this role. Evaluations, recommendations and training plans are connected through linked records, so employee, criterion, training and plan relationships can be followed inside one system.

The frontend does not expose secret keys on the client side. Airtable operations are handled through server-side routes. This approach improves security and helps control record creation and update operations.

**TABLE I. MAIN TECHNOLOGIES USED IN THE PROJECT**

| Layer          | Technology | Explanation                                                      |
|----------------|------------|------------------------------------------------------------------|
| Database       | Airtable   | Stores users, employees, evaluations, recommendations and plans. |
| Frontend       | Next.js    | Provides the web interface and protected application pages.      |
| Authentication | Clerk      | Handles login, registration, sessions and user identity.         |
| Deployment     | Vercel     | Publishes the frontend application on the web.                   |
| Automation     | n8n        | Runs background workflows for evaluation analysis.               |
| AI             | OpenRouter | Generates training recommendations from evaluation data.         |

## IV. DATABASE DESIGN

The Airtable base contains separate tables for each major part of the system. This makes the database easier to maintain and also prepares it for API usage from the Next.js frontend and n8n workflows. The tables are connected by linked record fields instead of being kept as unrelated text values.

The Users table stores login-related information such as name, e-mail, role, Clerk user ID and active status. The Employees table stores employee profile details such as name, e-mail, position, department, status and the linked user account. This connection allows the system to match the signed-in user with an employee profile.

The Evaluations table stores general evaluation records. The Evaluation Details table stores competency-based scores and comments for each evaluation. Competency Criteria defines the skills used during evaluation. Trainings contains the list of possible trainings. Training Recommendations stores suggested trainings, reasons, priorities and statuses. Training Plans and Training Plan Participants manage planned training activities and employee participation.

## V. USER ROLES AND AUTHORIZATION

The system uses role-based authorization to make the platform safer and easier to use. Each role sees only the pages and actions related to its responsibility. This prevents employees from reaching administrative pages and keeps human resources operations under control.

The Admin role has full access to the system. The HR role can manage employees, evaluations, recommendations and training plans. The Manager role can follow employees and review recommendations. The Evaluator role can create evaluations and review evaluation details. The Employee role can view personal information, personal recommendations and assigned training plans.

Authorization is applied both in the sidebar and at page level. Hidden pages are not only removed from the menu; restricted pages are also protected when a user tries to open them directly from the browser address bar. API routes are also prepared with permission checks for important create and update actions.

## **VI. USER INTERFACE**

The user interface is developed with Next.js. The design is simple, bright and understandable. The project includes a dashboard, employee management page, evaluation page, training recommendations page, training plans page, user management page, profile page, reports page and settings page.

The dashboard gives a quick summary of the system. Employee management lists employee records and helps authorized users inspect employee details. The evaluation page allows a new evaluation to be created with competency scores and comments. The recommendation page shows suggested trainings and allows approval or rejection.

The training plans page is used to create and manage training plans. It includes responsible person, training, dates, participants and status actions such as start, complete and cancel. The profile page is designed for employees. On this page, the logged-in employee can see personal information, evaluations, recommendations and training plans. The employee can also update training participation status with actions such as I will attend, I cannot attend, and I completed it.

## **VII. MAIN APPLICATION FLOW**

A typical usage scenario starts when an authorized user creates an employee evaluation. The evaluator selects the employee and enters scores for competency criteria. If a competency score is low, the system can create or later generate a training recommendation. The HR user can approve the recommendation and use it to create a training plan. Participants can be added to the plan, and employees can respond from their own profile page.

This flow connects evaluation, recommendation, planning and participation in a single structure. Therefore, the system does not only store data; it also supports the practical human resources process from analysis to training follow-up.

## **VIII. AUTOMATION AND ARTIFICIAL INTELLIGENCE**

The automation part of the project is prepared with n8n. The first necessary workflow is the AI Training Recommendation Creator. This workflow reads completed evaluations from Airtable, sends the evaluation details to OpenRouter, receives a JSON response and creates records in the Training Recommendations table.

The workflow searches for evaluation records whose automation status is pending, AI status is queued, API status is ready, evaluation status is completed and automation lock is not checked. After finding a record, the workflow locks it to prevent duplicate processing. Then it collects evaluation details, competency criteria, available trainings and existing recommendations.

OpenRouter is used as the AI provider. The AI model is instructed to return only valid JSON. Scores 1 and 2 are treated as high training need, score 3 is treated as medium need, and scores 4 and 5 usually do not require training. The AI must choose only from the existing training list and must not create duplicate recommendations for the same criterion.

After the AI response is parsed, n8n creates one training recommendation record for each valid recommendation. The evaluation record is then updated as completed and the AI summary, recommendation reasons and recommendation count are written back to Airtable. If an error occurs, the workflow marks the evaluation as failed and stores the error message in the automation error field.

## **IX. TESTING PROCESS**

The project was tested step by step during development. First, the Airtable base structure was prepared. Then the Next.js project was created and connected to Clerk for authentication. After that, protected pages, role-based menus and API permissions were tested.

Employee management, evaluation creation, recommendation management, training plan creation and employee profile functions were tested manually. The employee profile page was also tested by creating an employee record, joining a training plan and updating participation status.

The n8n workflow was prepared with a manual trigger before activating the schedule. This is important because the workflow should be tested safely before running automatically every five minutes. In the manual test, the workflow should find pending evaluations, create recommendations and update the evaluation automation status correctly.

## **X. RESULTS**

As a result of the project, a usable web-based training needs analysis platform was developed. The system supports secure login, role-based authorization, employee management, evaluation creation, training recommendation management, training plan management, employee self-service profile and reporting pages.

The project also includes an automation-ready structure. Important records contain fields such as n8n status, AI status, automation lock, automation error, AI summary and recommendation count. These fields make it possible to follow the automation process and understand whether a record was processed successfully.

## **XI. CONCLUSION**

This project shows how employee training needs can be managed through a structured digital system. By connecting evaluation scores with training recommendations, the platform helps human resources teams make more consistent and data-supported decisions. The role-based structure also makes the system suitable for different user groups such as admin, HR, manager, evaluator and employee.

Using Airtable, Next.js, Clerk, Vercel, n8n and OpenRouter together creates a flexible architecture. Airtable makes the database easy to manage during development, Next.js provides a modern frontend, Clerk handles authentication, and n8n with OpenRouter prepares the project for AI-supported automation. The system can be improved further and later migrated to a stronger backend such as Supabase if needed.

## **XII. FUTURE WORK**

Future work includes completing the remaining necessary n8n workflows. The second workflow can create a draft training plan when a recommendation is approved. The third workflow can follow training participation and reminders. The project can also be deployed on Vercel with production Clerk keys and tested with real users.

Another future improvement is improving the AI recommendation quality. More detailed prompts, better training matching rules and stronger reporting features can make the system more useful for human resources teams.

## **ACKNOWLEDGMENT**

This project was developed as a training needs analysis automation system by combining database design, web development, authentication, role management and workflow automation concepts.

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